

Joshua Jang

University of Toronto, Canada

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Research & Art Portfolio: musicofmusix.github.io

Research Interests

Computational Photography, Computer Graphics, Human-Computer Interaction for Creativity

Education

2020.09 ~ 2026.06 **University of Toronto**, Toronto, Canada
(Expected) *Honours Bachelor of Science in Computer Science and Cognitive Science*
Cumulative GPA: 3.90 / 4.00
Dean's List: Winter 2024, Winter 2025
Mandatory Military Service, Republic of Korea (2021.06 ~ 2022.12)

Research Experience

2025.09 ~ Present **University of Toronto**, Toronto, Canada
Undergraduate Research Assistant @ Dynamic Graphics Project (DGP) Lab
Supervisors: Prof. Karan Singh, Karran Pandey

- Contributing to the development of an interactive, intelligent image brush tool for rapid scene composition and editing.
- Researching user experience bottlenecks in existing image editing workflows, formulating examples and edge cases for method evaluation.
- Exploring image context representations across levels of abstraction, with combined features from vision foundation models and classical descriptors.

2025.01 ~ 2025.05 **Samsung Electronics**, Toronto, Canada
Research Intern @ AI Center - Toronto
Supervisors: Prof. Michael Brown, Dr. Hakki Karaimer

- Developed a compact, interpretable model of photographic film color for emulation using digital images (see **Publications**). Carried out the full project pipeline with advisory input from supervisors.
- Implemented utilities for camera image signal processing (ISP) analysis in Samsung's projects, notably a vectorized RAW renderer and a homography-based automated color chart patch extractor.
- Built RAW-JPG image datasets under various environments, lighting conditions, and ISPs, conducting capture, alignment, and post-processing.

Publications

2025 **Hyun Jo (Joshua) Jang**, Hakki Karaimer, Michael Brown
Emulating Emulsion: A Compact Physically-Based Model for Film Colour
ACM SIGGRAPH 2025 Posters

Teaching Experience

- 2025.09 ~ Present **University of Toronto**, Toronto, Canada.
Undergraduate Advisor @ Computer Science Student Union
- Mentoring a small cohort of undergraduates exploring artist-driven computational research, guiding them through weekly literature reviews, research design, and project development.
 - Facilitating group discussions and critique sessions to help students build confidence in pursuing interdisciplinary research ideas.

Exhibitions

- 2024 **Hart House Camera Club 103rd Annual Exhibition**, Toronto, Canada.
William J. Dowkes Award for People Photography, Winner
- 2023 **Interactive Media Exhibition for Children**, Gimpo, South Korea.
Exhibitor, Artist

Creative Projects

- 2023 **Interactive Virtual Subway Station**
- Designed and developed a stylized 3D subway station, with visuals replicating 1990s Japanese hand-drawn animation through custom-made non-photorealistic rendering (NPR) shaders in Unity (see **Portfolio**).
 - Installed as part of an in-person interactive exhibition for children, where attendees entering the building would be greeted with a virtual subway train arriving and departing towards the next showroom.
- 2022 **Aesthetics-Driven Interactive Poster/Game UI Demo**
- Developed an interactive poster, where the visual flair of motion graphics and the interactivity of video games were combined in the Unity engine (see **Portfolio**).
 - Implemented a fully animated humanoid character featuring facial expressions, real-time conditional shader replacement, and randomly generated relaxed cubic splines to achieve stylistic effects.
- 2016 ~ Present **Photography, Graphic Design, 3D Art & Web Development**
- Highlighted projects include photography, 3D art with procedurally generated displacement maps, graphic posters, and more (see **Portfolio**).
 - Designed and implemented a portfolio website in HTML and Tailwind CSS, inspired by the Swiss grid system principles in graphic design.

Skills

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|-------------|---|
| Programming | C/C++, C#, Python, Lua |
| Creativity | Photoshop, Illustrator, Lightroom, Unity, Blender, Cinema 4D, HTML, CSS |
| Languages | English (Fluent), Korean (Fluent) |